Worksheet 2.3 — Simplifying Rational Expressions

Grade 11 Functions, University (MCR3U) · Unit 2: Equivalent Algebraic Expressions

The Big Ideas

A **rational expression** is a ratio of polynomials. We simplify it by **factoring** and **cancelling common factors** — never terms — and we always record the **restrictions**: the values of x that make an *original* denominator zero. A simplified expression is equivalent to the original only for allowed values of x; where a cancelled factor is zero the graph has a **hole**.

- **Steps:** (1) factor completely; (2) write the restrictions from the *original* denominators; (3) cancel common factors; (4) keep the restrictions.
- **Cancel factors, not terms:** $\dfrac{x+6}{6} \neq x+1$ because the 6 is *added*.
- **Opposites:** $\dfrac{a-b}{b-a} = -1$. Rewrite $b-a = -(a-b)$ before cancelling.
- **Dividing** means multiplying by the reciprocal — and the divisor's *numerator* becomes a restriction too.
- **Hole vs asymptote:** a restriction whose factor cancels is a hole; a restriction that survives in the denominator is a vertical asymptote. Find the hole's y by substituting into the simplified form.

Concept Essentials

Cancel factors, not terms

A rational expression is simplified by factoring numerator and denominator and cancelling common **factors**. $\dfrac{x+6}{6}$ cannot be cancelled: the 6 is a *term*. $\dfrac{a-b}{b-a} = -1$.

Restrictions

Every value that makes an *original* denominator zero is excluded — and stays excluded after cancelling. When dividing, also exclude values that make the divisor's numerator zero.

Holes and asymptotes

A cancelled factor leaves a **hole**: substitute the restricted x into the simplified expression for its y. A factor that stays in the denominator gives a vertical **asymptote**.

Worked Examples

Example 1 — Restrictions

State the restrictions on $\dfrac{x+5}{x^2-3x-10}$.

Solution. Factor the denominator: $x^2-3x-10 = (x-5)(x+2)$. Set each factor to 0.
$x \neq -2,\ 5$. The numerator never creates a restriction.

Example 2 — Monomials

Simplify $\dfrac{20m^4n^2}{35mn^6}$.

Solution. Numbers: $\dfrac{20}{35} = \dfrac{4}{7}$. Variables: $m^{4-1} = m^3$ stays on top; $n^{2-6} = n^{-4}$, so n^4 goes to the bottom.

$= \dfrac{4m^3}{7n^4}$, with $m \neq 0,\ n \neq 0$.

Example 3 — Trinomial and difference of squares

Simplify $\dfrac{x^2-16}{x^2-x-12}$.

Solution. $\dfrac{(x-4)(x+4)}{(x-4)(x+3)}$; restrictions $x \neq -3,\ 4$. Cancel $(x-4)$.

$= \dfrac{(x+4)}{(x+3)}$, $x \neq -3,\ 4$.

Example 4 — Complex trinomial

Simplify $\dfrac{4x^2-9}{2x^2+x-6}$.

Solution. $4x^2-9 = (2x-3)(2x+3)$ and $2x^2+x-6 = (2x-3)(x+2)$.

Restrictions: $x \neq \frac{3}{2},\ -2$. Cancel $(2x-3)$.

$= \dfrac{(2x+3)}{(x+2)}$.

Example 5 — Opposite factors

Simplify $\dfrac{3-x}{x^2-9}$.

Solution. $3-x = -(x-3)$ and $x^2-9 = (x-3)(x+3)$.

$\dfrac{-(x-3)}{(x-3)(x+3)} = -\dfrac{1}{x+3}$, with $x \neq \pm 3$.

Example 6 — Multiplying

Simplify $\dfrac{x^2+2x-15}{x^2-9} \times \dfrac{x+3}{x+5}$.

Solution. Factor: $\dfrac{(x+5)(x-3)}{(x-3)(x+3)} \times \dfrac{x+3}{x+5}$. Restrictions: $x \neq -5,\ -3,\ 3$.

Everything cancels: $= 1$, $x \neq -5,\ -3,\ 3$ — the answer is 1 *except* at three missing points.

Example 7 — Dividing

Simplify $\dfrac{2x^2-8}{x^2+6x+9} \div \dfrac{x-2}{x+3}$.

Solution. Factor and flip the second fraction: $\dfrac{2(x-2)(x+2)}{(x+3)^2} \times \dfrac{x+3}{x-2}$.

Cancel $(x-2)$ and one $(x+3)$: $= \dfrac{2(x+2)}{(x+3)}$.

Restrictions: $x \neq -3$ (denominators) and $x \neq 2$ (we divided by $\frac{x-2}{x+3}$). So $x \neq -3,\ 2$.

Example 8 — Application: density

A block has mass $(15x^2-4x-4)$ g and volume $(3x-2)$ cm^3. Write its density as a simplified expression and evaluate at $x=2$.

Solution. Density $= \dfrac{\text{mass}}{\text{volume}} = \dfrac{(5x+2)(3x-2)}{3x-2} = 5x+2$ g/cm^3, $x \neq \frac{2}{3}$.

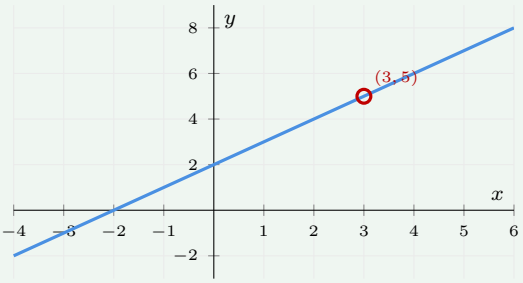
At $x=2$: 12 g/cm^3. Check: $\dfrac{60-8-4}{4} = 12$ ✓.

Example 9 — Finding a hole

Simplify $y = \dfrac{x^2-x-6}{x-3}$ and find the coordinates of the hole.

Solution. $\dfrac{(x-3)(x+2)}{x-3} = x+2$, $x \neq 3$.

At $x=3$ the line has $y=5$, so the hole is at $(3,5)$.



The graph is the line $y = x+2$ with one point removed.

Example 10 — Test before you trust

A student says $\dfrac{x^2+9}{x+3} = x+3$. Test the claim.

x	0	1	2	3
$\dfrac{x^2+9}{x+3}$	3	5/2	13/5	3
$x+3$	3	4	5	6
same?	✓	×	×	×

Solution. The values agree at $x = 0$ and $x = 3$ but not at $x = 1$ or $x = 2$. One mismatch is enough to prove the expressions are *not* equivalent. $x^2 + 9$ is a sum of squares and does not factor, so nothing cancels: the claim confuses a *term* with a *factor*.

Your Turn — Show Your Work

For each question, show all steps clearly in the space provided.

Question 1

State the restrictions on $\dfrac{2x-1}{x^2-x-12}$.

Question 2

Simplify $\dfrac{36a^2b^5}{48a^4b^2}$.

Question 3

Simplify $\dfrac{x^2-x-20}{x^2-25}$ and state the restrictions.

Question 4

Simplify $\dfrac{3x^2+x-2}{3x^2-5x+2}$ and state the restrictions.

Question 5

Simplify $\dfrac{2-x}{x^2-4x+4}$. (Look for opposites.)

Question 6

Simplify $\dfrac{x^2-1}{x^2+3x+2} \times \dfrac{x+2}{x-1}$. State all the restrictions.

Question 7

Simplify $\dfrac{x^2-6x+9}{x^2-4} \div \dfrac{x-3}{x+2}$. State all the restrictions.

Question 8

Simplify $\dfrac{2x^3-8x}{x^2-2x}$ and state the restrictions.

Question 9

The graph of $y = \dfrac{x^2 + x - 12}{x^2 - 9}$ has one hole and one vertical asymptote. Find the coordinates of the hole and the equation of the asymptote.

Question 10

A metal bar has mass $(10x^2 - x - 3)$ g and volume $(2x + 1)$ cm^3. Find its density in simplest form, and the density when $x = 4$.

Question 11

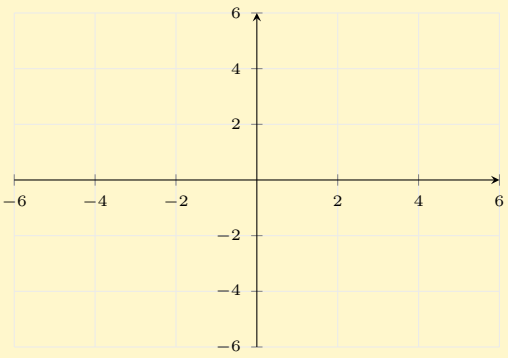
A student writes $\dfrac{x^2 + x - 6}{x^2 - 9} = \dfrac{x - 6}{-9}$, *cancelling the* x^2 *terms.* Test the claim at $x = 4$, explain the mistake, and simplify correctly.

Question 12

Explain why $\dfrac{x+4}{4}$ is not equal to $x+1$. Use the value $x=4$ and the words *term* and *factor*.

Question 13

Simplify $\dfrac{x^2-4}{x+2}$, then sketch its graph, marking the hole.



Question 14 — Challenge

Simplify $\dfrac{x^4-5x^2+4}{x^3-x^2-4x+4}$ and list all restrictions. (Factor the denominator by grouping.)

Question Recap & Answer Key

Question Recap

1. State the restrictions on $\dfrac{2x-1}{x^2-x-12}$.
2. Simplify $\dfrac{36a^2b^5}{48a^4b^2}$.
3. Simplify $\dfrac{x^2-x-20}{x^2-25}$ and state the restrictions.
4. Simplify $\dfrac{3x^2+x-2}{3x^2-5x+2}$ and state the restrictions.
5. Simplify $\dfrac{2-x}{x^2-4x+4}$. (Look for opposites.)
6. Simplify $\dfrac{x^2-1}{x^2+3x+2}\times\dfrac{x+2}{x-1}$. State all the restrictions.
7. Simplify $\dfrac{x^2-6x+9}{x^2-4}\div\dfrac{x-3}{x+2}$. State all the restrictions.
8. Simplify $\dfrac{2x^3-8x}{x^2-2x}$ and state the restrictions.
9. The graph of $y=\dfrac{x^2+x-12}{x^2-9}$ has one hole and one vertical asymptote. Find the coordinates of the hole and the equation of the asymptote.
10. A metal bar has mass $(10x^2-x-3)$ g and volume $(2x+1)$ cm^3. Find its density in simplest form, and the density when $x=4$.
11. A student writes $\dfrac{x^2+x-6}{x^2-9}=\dfrac{x-6}{-9}$, *cancelling the x^2 terms*. Test the claim at $x=4$, explain the mistake, and simplify correctly.
12. Explain why $\dfrac{x+4}{4}$ is not equal to $x+1$. Use the value $x=4$ and the words *term* and *factor*.
13. Simplify $\dfrac{x^2-4}{x+2}$, then sketch its graph, marking the hole.
14. Simplify $\dfrac{x^4-5x^2+4}{x^3-x^2-4x+4}$ and list all restrictions. (Factor the denominator by grouping.)

✓ Answer Key

1. $x^2-x-12=(x-4)(x+3)$, so $x\neq -3,\ 4$.
2. $\dfrac{3b^3}{4a^2}$, with $a\neq 0,\ b\neq 0$.
3. $\dfrac{(x-5)(x+4)}{(x-5)(x+5)}=\dfrac{(x+4)}{(x+5)}$, $x\neq -5,\ 5$.
4. $\dfrac{(3x-2)(x+1)}{(3x-2)(x-1)}=\dfrac{(x+1)}{(x-1)}$, $x\neq \frac{2}{3},\ 1$.
5. $\dfrac{-(x-2)}{(x-2)^2}=-\dfrac{1}{x-2}$, $x\neq 2$.
6. $\dfrac{(x-1)(x+1)}{(x+1)(x+2)}\times\dfrac{x+2}{x-1}=1$, $x\neq -2,\ -1,\ 1$.
7. $\dfrac{(x-3)^2}{(x-2)(x+2)}\times\dfrac{x+2}{x-3}=\dfrac{(x-3)}{(x-2)}$, $x\neq -2,\ 2,\ 3$.
8. $\dfrac{2x(x-2)(x+2)}{x(x-2)}=2(x+2)$, $x\neq 0,\ 2$.
9. $\dfrac{(x+4)(x-3)}{(x-3)(x+3)}=\dfrac{x+4}{x+3}$. Hole: $x=3$, $y=\frac{7}{6}$, at $(3,\frac{7}{6})$. The factor $(x+3)$ survives, so the vertical asymptote is $x=-3$.
10. $\dfrac{(5x-3)(2x+1)}{2x+1}=5x-3$ g/cm^3, $x\neq -\frac{1}{2}$. At $x=4$: 17 g/cm^3.
11. At $x=4$: $\dfrac{16+4-6}{16-9}=2$ but $\dfrac{4-6}{-9}=\frac{2}{9}$. The student cancelled x^2, which is a *term* in each part. Correct: $\dfrac{(x-2)(x+3)}{(x-3)(x+3)}=\dfrac{x-2}{x-3}$, $x\neq\pm 3$ (at $x=4$ this gives 2 ✓).
12. At $x=4$: $\dfrac{8}{4}=2$ but $x+1=5$. The 4 in the numerator is *added* to x (a term), so it cannot be cancelled; only a *factor* that multiplies the whole numerator and the whole denominator can be cancelled.
13. $\dfrac{(x-2)(x+2)}{x+2}=x-2$, $x\neq -2$. The graph is the line $y=x-2$ with a hole at $(-2,-4)$.
14. Denominator: $x^2(x-1)-4(x-1)=(x-1)(x-2)(x+2)$. Numerator: $(x^2-1)(x^2-4)=(x-1)(x+1)(x-2)(x+2)$. Result $(x+1)$, $x\neq -2,\ 1,\ 2$.